10.11 From LSZ to the Feynman Rules

LecNote9 runs one example through the LSZ formula, the φ^3 four-point function at tree level (pp. 1–5), and then states ten rules for iT at any order (pp. 6–7). The questions sit in the seams:

- "Only one disconnected product on p. 2 is checked, and 'no scattering' sounds like 'we don't care'. Where is the amplitude for nothing happening?"
- "We did one tree. Why do the rules hold for every diagram, loops included, and why is exactly one $(2\pi)^4\delta^4$ left (p. 5)?"
- "A loop on an external leg puts $-i/(k^2+m^2) = 1/0$ next to the on-shell line. Why is each external line just 1 (rule 6)?"
- "What are Z_g, A and B doing in the rules, and why iT?"

In short: every step of the example is local to one line or one vertex, so it repeats in any diagram. Disconnected terms need forward kinematics, and on shell they drop out of the LSZ integral altogether. The external factor 1 is exact, because LSZ fixes the mass and the normalization of the field, and A, B enforce them. Translation invariance leaves one δ^4. Every tree is i times a real number, so T is real at tree level.
Conventions are Srednicki's: metric $(-,+,+,+)$, $k^2 = -m^2$ on shell, $\Delta(x) = \int \frac{d^4k}{(2\pi)^4} e^{ikx}\tilde{\Delta}(k)$, $\tilde{\Delta}(k) = 1/(k^2+m^2-i\epsilon)$, $(-\partial^2+m^2)\Delta = \delta^4$, and $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \frac{1}{i}\Delta(x-y)$ in the free theory (LecNote9 writes e^{-ikx}; $\tilde{\Delta}$ is even, so nothing changes). Bold $\mathbf{\Delta}$ is the exact propagator. The interaction is $\mathcal{L}_1 = \frac{1}{6}Z_g g\varphi^3 + \mathcal{L}_{\rm ct}$, $\mathcal{L}_{\rm ct} = -\frac{1}{2}A\,\partial\varphi\partial\varphi - \frac{1}{2}Bm^2\varphi^2 + Y\varphi$, with $A = Z_\varphi - 1$, $B = Z_m - 1$.

Connected correlators and the disconnected terms

With $\delta_j \equiv \frac{1}{i}\,\delta/\delta J(x_j)$ and $Z = e^{iW}$, where iW is the sum of connected diagrams (LecNote8 pp. 8–9), the connected correlators are (p. 3)

$$\begin{aligned} \langle 0|T\varphi(x_1)\cdots\varphi(x_E)|0\rangle_C &\equiv \delta_1\cdots\delta_E\, iW|_{J=0}, \\ \frac{\delta_1\delta_2 Z}{Z} &= \delta_1\delta_2\, iW + (\delta_1 iW)(\delta_2 iW), \end{aligned} \tag{10.11.1}$$

with a plus sign in the chain rule (p. 1's minus is harmless: $\delta_j iW|_{J=0}$ is $\langle 0|\varphi|0\rangle = 0$). With $\mathbf{\Delta}_{ij} \equiv \mathbf{\Delta}(x_i - x_j)$,

$$\langle 0|T\varphi_1\varphi_2\varphi_3\varphi_4|0\rangle = \langle\cdots\rangle_C - \mathbf{\Delta}_{12}\mathbf{\Delta}_{34} - \mathbf{\Delta}_{13}\mathbf{\Delta}_{24} - \mathbf{\Delta}_{14}\mathbf{\Delta}_{23}, \tag{10.11.2}$$

the -1 being $(1/i)^2$. In general the full correlator is the sum over all ways of splitting the points into groups of at least two, each group contributing its connected correlator.

LSZ (LecNote9 p. 2) puts $i\int d^4x\, e^{ikx}(-\partial^2+m^2)$ on each incoming point and the same with $e^{-ik'x'}$ on each outgoing one. Integrating by parts, it says: Fourier transform the correlator, multiply by $i(k^2+m^2)$ for each external particle, and go on shell. In (10.11.2) the pairing $(12)(1'2')$ comes with $(2\pi)^4\delta^4(k_1+k_2)$, which vanishes because $k_1^0+k_2^0>0$. In $(11')(22')$ each pair gives

$$i(k_1^2+m^2)\,i(k_1'^2+m^2)\,\tfrac{1}{i}\,(2\pi)^4\delta^4(k_1-k_1')\,\tilde{\mathbf{\Delta}}(k_1)=i(2\pi)^4\delta^4(k_1-k_1')\,\tilde F(k_1), \quad (10.11.3)$$

with p. 2's $\tilde F(k)=(k^2+m^2)^2\tilde{\mathbf{\Delta}}(k)$. This and $(12')(21')$ need some $k'=k$: nothing is exchanged, p. 2's "no scattering", and taking every $k'\neq$ every k (LecNote4 p. 3) removes them. Moreover, since $\tilde{\mathbf{\Delta}}$ has a simple pole at $k^2=-m^2$ (shown below), $\tilde F\to 0$ on shell: even in forward kinematics the term is $0\cdot\delta^4$. The plane-wave LSZ integral does not contain the amplitude for nothing happening. That amplitude, the "1" of S, is the overlap terms $O_{1'1}O_{2'2}+O_{1'2}O_{2'1}$ of (10.6.18) (§10.6), dropped at the time-ordering step on LecNote4 p. 5.

For $n\to n'$ each group in a product brings a $(2\pi)^4\delta^4$ of its own momenta. A group of only incoming (or only outgoing) particles would need zero energy; one particle against $b\geq 2$ is a stable particle decaying or being formed, impossible since b on-shell particles have invariant mass at least bm; one against one is forward, with $\tilde F\to 0$. What remains are independent subprocesses with at least two particles on each side, such as two pairs scattering separately in $4\to 4$ (cluster decomposition). Each needs a subset of the incoming momenta to have the same total as a subset of the outgoing ones. For momenta in general position, where none does, only the fully connected correlator survives:

$$\langle f|i\rangle=(2\pi)^4\delta^4\Big(\sum k_{\text{in}}-\sum k_{\text{out}}\Big)\,iT, \qquad S=1+iT. \quad (10.11.4)$$

A CONNECTED DIAGRAM IN MOMENTUM SPACE

Take a connected diagram with V vertices, I internal and E external lines, for now with no loop on an external line. In position space it is $(iZ_g g)^V S^{-1}\int\prod_v d^4y_v\prod_{\text{lines}}\frac{1}{i}\Delta$. The $3!$ ways of attaching three distinct lines to a vertex $\frac{i}{6}Z_g g\int d^4y\,(\delta/i\delta J)^3$ cancel the $\frac{1}{6}$; each line is $(\frac{1}{i})^2\cdot i\Delta=\frac{1}{i}\Delta$ (p. 4). S counts the symmetries that leave every external line in place: 1 for every tree (p. 3), possibly more with loops (rule 8; §10.9). Three local steps take this to momentum space.

External lines. $i\int d^4x\,e^{ikx}(-\partial_x^2+m^2)\frac{1}{i}\Delta(x-y)=e^{iky}$: an incoming particle gives the factor 1 and feeds k into its vertex; an outgoing one gives $e^{-ik'y}$ and takes k' out.

Internal lines. With $\Delta(y-z)=\int \frac{d^4q}{(2\pi)^4}e^{iq(y-z)}\tilde{\Delta}(q)$ each internal line keeps $\int d^4q/(2\pi)^4$ and $-i/(q^2+m^2-i\epsilon)$, with q flowing from z to y. Reversing the arrow is $q\to -q$, which changes nothing: rule 3's "arbitrarily".
Vertices. Each y_v now appears only in exponentials, and $\int d^4y$ gives $(2\pi)^4\delta^4$ of the momentum flowing in minus that flowing out (rule 5).
Counting. Adding the arguments of the V delta functions, each internal momentum cancels, so together they contain $\delta^4(\sum k_{\text{in}}-\sum k_{\text{out}})$. The other $V-1$ fix $V-1$ internal momenta, and

$$L = I - V + 1 \tag{10.11.5}$$

stay free, one per closed loop (rule 7). Of the factors $(2\pi)^{4V}$ and $(2\pi)^{-4I}$, $V-1$ pairs go with the δ's used up, each loop keeps $d^4l/(2\pi)^4$, and one $(2\pi)^4\delta^4$ is left:

$$\text{diagram} = (2\pi)^4\delta^4\Big(\sum k_{\text{in}}-\sum k_{\text{out}}\Big) \times \frac{(iZ_g g)^V}{S}\prod_{\text{loops}}\int\frac{d^4l}{(2\pi)^4}\prod_{\text{internal}}\frac{-i}{q^2+m^2-i\epsilon}. \tag{10.11.6}$$

The single δ^4 is not an accident of perturbation theory. The connected correlator is unchanged when all x_j shift by the same a, so its LSZ transform equals itself times $e^{ia(\sum k_{\text{in}}-\sum k_{\text{out}})}$ for every a and must be proportional to $\delta^4(\sum k_{\text{in}}-\sum k_{\text{out}})$. Stripping off $(2\pi)^4\delta^4$ defines iT (p. 5); LecNote10 squares this δ^4.

External lines: amputation and the residue

Beyond tree level the line at an external point can carry self-energy loops, and then a propagator $-i/(k^2+m^2)$ with the on-shell momentum appears: $1/0$. Sum all insertions on the leg. With $i\Pi(k^2)$ the one-particle-irreducible two-point diagrams, counterterm vertex included (LecNote12 pp. 1–2),

$$\frac{1}{i}\tilde{\Delta}+\frac{1}{i}\tilde{\Delta}\,(i\Pi)\,\frac{1}{i}\tilde{\Delta}+\cdots=\frac{1}{i}\,\frac{1}{k^2+m^2-i\epsilon-\Pi(k^2)}\equiv\frac{1}{i}\tilde{\mathbf{\Delta}}(k). \tag{10.11.7}$$

So the sum of connected diagrams is the sum of *amputated* ones, with no corrections on external lines, times one exact propagator per leg, and on each leg

$$i(k^2+m^2)\,\frac{1}{i}\tilde{\mathbf{\Delta}}(k)=\frac{k^2+m^2}{k^2+m^2-\Pi(k^2)}\;\longrightarrow\;1\qquad(k^2\to -m^2) \tag{10.11.8}$$

if and only if $\Pi(-m^2)=\Pi'(-m^2)=0$ (LecNote12 p. 3). These are LSZ's own assumptions: in the Lehmann–Källén form (LecNote11) the one-particle pole of $\tilde{\mathbf{\Delta}}$ sits at the physical mass with residue $|\langle k|\varphi(0)|0\rangle|^2$, and LSZ

takes that mass as the m in $-\partial^2 + m^2$ and $\langle k|\varphi(0)|0\rangle = 1$ (§10.6). So rule 6's factor 1 is exact at every order, and loops on external lines are never drawn: they are already in the 1. If a condition failed:

- With $\Pi'(-m^2) \neq 0$ each leg gives the residue $R = 1/[1 - \Pi'(-m^2)]$. The normalized field $\varphi/\sqrt{R}$ has E-point function $R^{-E/2}$ times that of φ, so the amplitude is $(\sqrt{R})^E$ times the amputated diagrams.
- With $\Pi(-m^2) \neq 0$ the pole is elsewhere and (10.11.8) goes to 0: the "wrong mass" of §10.6.
- One insertion alone gives $i(k^2+m^2)\frac{1}{i}\tilde{\Delta}(i\Pi)\frac{1}{i}\tilde{\Delta} = \Pi/(k^2+m^2)$: the naive $1/0$ is $0/0$, and with $\Pi = O\big((k^2+m^2)^2\big)$ it vanishes on shell.

Counterterms, tadpoles and the coupling

With $\partial \to ik$ the quadratic counterterms in the action become

$$-\frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\,\tilde{\varphi}(-k)\,(Ak^2 + Bm^2)\,\tilde{\varphi}(k), \qquad \text{vertex:}\quad i \times \big(-\tfrac{1}{2}\big)(Ak^2 + Bm^2) \times 2 = -i(Ak^2 + Bm^2), \tag{10.11.9}$$

rule 9's value: a vertex is i times its coefficient in $\mathcal{L}$ times the number of ways to attach its lines (3! for φ^3, here 2). On an external leg it is part of Π, hence of the factor 1; on an internal line it is a diagram of its own. Of the four conditions on LecNote8 p. 1, two (physical m, unit residue) are $\Pi(-m^2) = \Pi'(-m^2) = 0$ and fix A and B. The third, $\langle 0|\varphi|0\rangle = 0$, fixes Y to cancel the sum of all tadpoles, pieces with no external line joined to the rest of a diagram by a single line (LecNote8 p. 13; §10.10), so neither appears in the rules. The fourth is not an LSZ condition: Z_g defines g, through a condition such as $V_3(0,0,0) = g$ on the exact vertex (LecNote13 p. 4). Since $Z_g = 1 + O(g^2)$, the tree vertex is ig.

Phase and order

By (10.11.6) a tree ($L = 0$, $I = V - 1$) is $(ig)^V(-i)^{V-1} = ig^V$ times real denominators: T is real at tree level wherever no denominator vanishes, which is why $S = 1 + iT$ carries the i. After Feynman parameters and the Wick rotation $d^4l = i\,d^4\bar{l}$ (LecNote12 pp. 4–6) each loop adds one more i, and $i^V(-i)^I i^L = i^{V-I+L} = i$ again; but now a denominator can change sign inside the integration region, and the $i\epsilon$ gives T an imaginary part, for $2 \to 2$ once $s > 4m^2$ (LecNote12 p. 15, LecNote14 p. 8). Each vertex has three line ends and each internal line uses two, so $3V = 2I + E$, and with (10.11.5)

$$V = E - 2 + 2L. \tag{10.11.10}$$

At fixed E each loop costs g^2: the loop expansion is an expansion in g^2. For the same reason A, B and $Z_g - 1$ start at $O(g^2)$: the one-loop Π ($E = 2$, $V = 2$) and vertex ($E = 3$, $V = 3$) they cancel are g^2 beyond tree level.

A WORKED CHECK: $2 \to 2$

At tree level $V = 2$, $I = 1$: three labelled diagrams, one per partner of particle 1 at its vertex (p. 3's three groups), each with $S = 1$, and conservation fixes the internal momentum to $k_1 + k_2$, $k_1 - k_1'$ or $k_1 - k_2'$ (p. 8). The rules give

$$iT = (ig)^2\Big[\frac{-i}{(k_1+k_2)^2+m^2} + \frac{-i}{(k_1-k_1')^2+m^2} + \frac{-i}{(k_1-k_2')^2+m^2}\Big], \qquad T = g^2\Big[\frac{1}{m^2-s} + \frac{1}{m^2-t} + \frac{1}{m^2-u}\Big], \tag{10.11.11}$$

with $s = -(k_1+k_2)^2$, $t = -(k_1-k_1')^2$, $u = -(k_1-k_2')^2$, as on p. 5 and p. 8. On shell $s+t+u = 4m^2$, and physically $s \geq 4m^2$, $t, u \leq 0$. So $m^2 - s \leq -3m^2$ and $m^2 - t$, $m^2 - u \geq m^2$: no denominator vanishes and T is real. At threshold ($s = 4m^2$, $t = u = 0$) $T = 5g^2/3m^2$; T is symmetric in s, t, u.

At one loop ($V = 4$, $I = 4$) the table below lists the labelled diagrams. To the three kept topologies (their pieces are computed in LecNote12–13 and assembled in LecNote14) add the counterterm (10.11.9) on the internal line and the $O(g^4)$ part of $(iZ_g g)^2$ in the tree diagrams. As a check on the counting, $1/S$ summed over all 39 labelled diagrams is 24, what φ^3 theory in zero dimensions gives for this order of the connected four-point function.

diagram	number	S
box	3	1
triangle at one vertex	6	1
bubble on the internal line	3	2
not drawn: bubble on an external line	12	2
not drawn: tadpole on the internal line	3	2
not drawn: tadpole on an external line	12	2

ONE-LOOP $2 \to 2$ DIAGRAMS IN φ^3

External lines labelled $1, 2, 1', 2'$; S counts the symmetries that fix them. External bubbles are in the factor 1; Y cancels the tadpoles.